Noise Event Detection in Real-World Indic Speech: Team Deadlock — Track 1 (Datathon@IndoML 2026)

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***Abstract*—We present Team Deadlock's system for Track 1 (Noise Event Detection) of Datathon@IndoML 2026, which asks participants to localize non-stationary background noise events – vehicle horns, dogs barking, doorbells, kitchen appliances – in real-world Indic speech, with precise onset/offset timestamps. Our primary system is a compact convolutional-recurrent neural network (CRNN) trained on log-mel spectrogram features from the Gold tier of the Vaani-Noise-Event-Dataset. We additionally explored a Wav2Vec2-based detector and designed (but did not complete in time) a weak-supervision scheme to exploit the untimestamped Bronze tier. Our CRNN submission scores 0.9221 (Combined Event-F1 + Segment-Dice) on the public leaderboard, ranked 51/188 as of the final submission.**

I. Introduction

Speech recordings collected in real Indian environments are dominated by bursty, non-stationary background events – a passing motorbike, a hen, a doorbell, devotional music – that degrade downstream Automatic Speech Recognition (ASR), and that standard sound-event-detection benchmarks (AudioSet, DESED, DCASE) under-represent. Track 1 of Datathon@IndoML 2026 asks participants to detect such events in the Vaani corpus, a large-scale, multi-language Indian speech dataset, and output their onset/offset timestamps. We treat this as a frame-level binary detection problem: predict a per-frame noise-presence probability and decode it into discrete events.

II. Training Data and Preprocessing

The organizers provide ∼150 hours of training audio at three annotation quality tiers: **Gold** (∼20h, 9,426 clips, mutually verified onset/offset timestamps), **Silver** (∼100h, timestamps present but unverified), and **Bronze** (∼30h, only clip-level noise tags, no timestamps).

Our primary (submitted) model was trained on the **Gold tier only** (9,426 clips), held out 2,501 clips for local validation. Audio is resampled to 16 kHz mono and converted to a 64-band log-mel spectrogram (1024-point FFT, 512-sample hop, ≈32ms frames at this hop/SR). No data augmentation was applied to the submitted model. No external datasets were used – only the organizer-provided Vaani corpus.

We also designed (but did not complete a full run of) a weak-supervision scheme for the Bronze tier: the trained CRNN is run over every Bronze clip, and a clip is accepted as a pseudo-labeled training example only when the model's own prediction agrees with the clip-level tag (noise tag present and the model confidently predicts at least one event, mean positive-frame probability ≥ 0.6; or no tag and no predicted event). Disagreements are discarded. This is implemented in `src/bronze_weak_supervision.py` but was not run to completion before the deadline.

III. Proposed Method

A. Architecture — CRNN (submitted model)

The submitted model is a small convolutional-recurrent network:

- **Input:** 64-band log-mel spectrogram.
- **CNN backbone:** two Conv2D blocks ($16 \rightarrow 32$ channels, 3×3 kernels, batch normalization, ReLU, frequency-axis max pooling).
- **Temporal modeling:** single-layer bidirectional LSTM (hidden size 64) over the CNN's frame sequence.
- **Head:** a linear layer producing one logit per frame, passed through a sigmoid to obtain a noise-presence probability.

B. Event Decoding

Frame-level probabilities are converted to discrete onset/offset events via: probability threshold (0.35) → temporal median filter (window 7, to remove isolated spurious frames) → minimum-duration filter (0.15s) → merge-gap filter (0.10s, to join adjacent detections into single events).

C. Training

Optimizer: Adam, learning rate 1×10^{-3}, loss: `BCEWithLogitsLoss` on per-frame binary noise targets, trained for 5 epochs with batch size 4 on the 9,426-clip Gold split. Environment: NVIDIA RTX 4050 Laptop GPU (6GB VRAM), Python 3.12.7, PyTorch 2.6.0 (CUDA 12.4), Windows 11.

D. Alternative: Wav2Vec2-based detector (not submitted)

We also fine-tuned a detector built on a frozen `facebook/wav2vec2-base` encoder (Hugging Face) with a BiGRU head over its native ∼50fps frame rate, trained with `BCEWithLogitsLoss` using a `pos_weight` computed from the frame-level class imbalance (an earlier run without this weighting collapsed to near-constant predictions). On a

5,000-clip Gold subset this scored Combined 0.7408 (Event-F1 0.1811, Dice 0.5597) – below the CRNN's 0.8418 local-validation score – so it was kept as a documented experiment rather than submitted. Code: `src/advanced_model.py`, `src/train_advanced.py`.

IV. Results and Conclusion

TABLE I
Track 1 Results

Model	Split	Combined
CRNN (Gold-only, submitted)	Local val (2,501 clips)	0.8418
CRNN (Gold-only, submitted)	Leaderboard (public, 11h)	**0.9221**
CRNN (Gold + 5k Silver)	Local val	0.8436
CRNN (Gold + 10k Silver)	Local val	0.7906
Wav2Vec2 + BiGRU (5k Gold)	Local val	0.7408

Our CRNN submission achieved a public leaderboard Combined score of **0.9221** (Event-F1 0.3014, Segment Dice 0.6206), placing **51st of 188** participating teams. The leaderboard score exceeds our local validation estimate, which we attribute to differences between our held-out validation split and the organizers' withheld evaluation distribution. The Wav2Vec2 alternative underperformed the CRNN baseline at the data scale we were able to train it at, likely due to the small clip-level batch size our 6GB-VRAM hardware budget allowed; a full 9,426-clip run was planned but not completed before the deadline.

A. Limitations and Future Work

Our submission uses only the Gold tier. Adding 5k Silver clips gave a marginal local gain (0.8436 vs. 0.8418) while 10k Silver clips hurt (0.7906), suggesting the unverified Silver timestamps need filtering or noise-robust training; we did not submit these variants. Completing the Bronze-tier weak-supervision pipeline described in Section II is another direct path to improvement that was designed but not finished in time. A full-scale Wav2Vec2 run and a CRNN/Wav2Vec2 ensemble are natural next steps.

References

References

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[3] A. Baevski, H. Zhou, A. Mohamed, M. Auli, "wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations," https://huggingface.co/facebook/wav2vec2-base.